

## **Analysis of the Liverpool Smart Pedestrians Project**

### **A. Introduction and Objectives**

The Liverpool Smart Pedestrians project aimed to develop a privacy-compliant, real-time traffic monitoring system using edge-computing and computer vision technologies. The primary objective was to collect accurate pedestrian, vehicle, and cyclist movement data to support data-driven urban planning decisions in Liverpool, New South Wales. By leveraging existing CCTV infrastructure, the project sought to minimize deployment costs while increasing the value of already-installed city assets.

The project addressed several urban planning challenges, including rapid population growth, increasing pedestrian congestion, and the need for sustainable mobility planning. Traditional traffic monitoring methods often require intrusive infrastructure or raise privacy concerns. This project aimed to overcome these issues by processing video data locally at the edge, ensuring that no raw video or personal data left the CCTV network.

### **B. Methodology**

The methodology centered on designing and evaluating an edge-computing visual sensor capable of detecting and tracking multiple transportation modes in real time. Community workshops and consultations with urban planners were conducted early in the project to define requirements such as privacy compliance, multi-modal detection, scalability, and interoperability.

Key constraints during sensor development included limited network bandwidth, strict privacy regulations, and the need to operate reliably in outdoor environments. These constraints informed the choice of onboard processing, metadata-only transmission, and robust hardware packaging.

### **C. Technology and Implementation**

The sensor hardware was built around the NVIDIA Jetson TX2, selected for its balance of computational power, energy efficiency, and GPU acceleration capabilities. The Jetson TX2 enabled real-time execution of the YOLO V3 deep learning algorithm for object detection, which was chosen due to its strong trade-off between accuracy and speed.

YOLO V3 was combined with the SORT tracking algorithm to maintain object identities across frames. The edge-computing paradigm allowed video analytics to be performed locally, reducing bandwidth usage, improving response time, and enhancing privacy. This aligns closely with Module 2 principles, which emphasize decentralized processing, low latency, and data minimization.

## **D. Validation and Performance**

Validation experiments were conducted using the Oxford Town Center dataset and real-world CCTV deployments. Results showed an average accuracy of approximately 69% for pedestrian detection, with performance reaching nearly 20 frames per second.

System utilization metrics demonstrated stable memory and temperature levels, while GPU usage occasionally dropped due to CPU-based tracking becoming a bottleneck. Despite this, the sensor successfully captured traffic trends and flow patterns, which are often more valuable for urban planning than exact object counts.

## **E. Real-World Applications**

Two real-world deployments demonstrated the system's effectiveness. The indoor deployment during an emergency evacuation successfully detected abnormal pedestrian movement patterns, showing potential for safety monitoring applications.

The outdoor deployment in Liverpool provided continuous, multi-day traffic data that revealed daily and weekly mobility patterns. These insights can directly inform urban planning decisions such as crosswalk placement, traffic signal timing, and pedestrian infrastructure improvements. This application strongly connects to Module 3 concepts involving sensor networks and edge gateways.

## **F. Challenges and Future Work**

Major challenges included detection accuracy in dense crowds, underutilization of GPU resources, and tracking limitations during occlusions. The authors proposed solutions such as GPU-accelerated tracking, improved deep learning frameworks, and upgrading to newer hardware platforms like NVIDIA Xavier.

Since 2019, advancements in edge AI hardware, newer object detection models (e.g., YOLOv7, YOLOv8), and improved communication protocols such as 5G and edge-to-cloud orchestration could significantly enhance similar systems today.

## **G. Personal Evaluation**

Overall, the Liverpool Smart Pedestrians project was a successful demonstration of how edge computing and AI can support smart city initiatives. Its strong focus on privacy, scalability, and real-world deployment makes it highly relevant for modern urban planning. Future improvements could include adaptive model retraining, multi-camera sensor fusion, and integration with predictive traffic analytics to further enhance decision-making capabilities.

## References

Barthélemy, J., Verstaevel, N., Forehead, H., & Perez, P. (2019). *Edge-computing video analytics for real-time traffic monitoring in a smart city*. **Sensors**, **19**(9), 2048.  
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